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Evaluation of a WeChat-based dementia-specific training program for nurses in primary care settings: A randomized controlled trial

Feilong Wang[a], Lily Dongxia Xiao[b], Kaifa Wang[c], Min Li[d], Yanni Yang[a,*]

[a] School of Nursing, Third Military Medical University, No.30 Gaotanyan Street, Chongqing 400038, People's Republic of China
[b] School of Nursing and Midwifery, Finders University, Adelaide, SA 5001, South, Australia
[c] Department of Mathematics, School of Biomedical Engineering, Third Military Medical University, No.30 Gaotanyan Street, Chongqing 400038, People's Republic of China
[d] School of Psychology, Third Military Medical University, No.30 Gaotanyan Street, Chongqing 400038, People's Republic of China

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ABSTRACT

Community nurses play a crucial role in early detection and timely diagnosis of dementia. However, they are usually not prepared for the role through their formal education, particularly in low- and middle-income countries due to undeveloped nursing curriculum in dementia care. This paper describes a two-arm cluster-randomized controlled trial to improve community nurses' knowledge, attitudes, and practice changes using an innovative and interactive mobile phone applet-based activity in primary care settings. The intervention sites received dementia-specific training and control sites received care training for older people with disability. Both groups completed measures assessing dementia knowledge, attitudes, and intentions to make changes to achieve early detection and a timely diagnosis of dementia immediately after training and at 3-month follow-up. The intervention group provided feedback immediately after training and at 3-month follow-up. The main results show that the intervention group demonstrated significant improvement in dementia knowledge and attitudes from baseline immediately after training and at the 3-month follow-up. The intervention group also showed more intentions to make changes to achieve early detection of dementia. Feedback suggested the program was well-received. Overall, the program showed acceptability and feasibility in improving nurses' dementia knowledge, attitudes, and intentions to achieve early detection of dementia.

1. Introduction

Dementia has become a global public health issue since the population is aging. In 2015, almost 46.8 million people worldwide lived with dementia, and this number is expected to reach 131.5 million by 2050 (Alzheimer's Disease International, 2015). Among those living with dementia, 58% are living in low- and middle-income countries (LMICs), and 94% of those live in their homes and are cared for by family members (Alzheimer's Disease International, 2015). Community nurses in LMICs play a vital role in caring for the population with dementia and supporting their caregivers. However, the primary healthcare system is largely undeveloped in LMICs. Although numerous reasons contribute to this undesirable situation, one of them is the lack of education and training for community nurses in dementia care in LMICs (Wang, Xiao, He, & DeBellis, 2014). China shares a large proportion of the global population living with dementia. Studies addressing community nurses' capability to improve dementia care are needed to produce research evidence that informs practice. This paper reports an innovative dementia-specific training program for nurses in primary care settings using WeChat, a simple mobile applet, to facilitate learner-centred interactive learning and enable changes in practice to achieve early detection and a timely diagnosis of dementia.

Community nurses are considered a key group in the primary care team that are responsible for a timely diagnosis of dementia, providing referrals, and risk reduction (Kallumpuram et al., 2015; Lee et al., 2010). They have more routine contact with residents and are in an ideal position to detect cognitive changes, identify needs, and provide subsequent management and coordinating services. However, only 38% of nurses have received dementia training in high-income countries (HICs) (Millard, Kennedy, & Baune, 2011). The situation is estimated much worse in LMICs considering the undeveloped dementia education in formal education for health professionals (Wang, Xiao, Ullah, He, & De Bellis, 2017). Studies have reported that nurses have difficulties identifying cognitive impairment and performing cognitive screening for people with subjective memory complaints (Trickey, Turton, Harvey, Wilcock, & Sharp, 2000). They also showed a lack of

* Corresponding author.
E-mail addresses: lily.xiao@finders.edu.au (L.D. Xiao), limin52267@tmmu.edu.cn (M. Li), yangyanni@tmmu.edu.cn (Y. Yang).

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knowledge in dementia risk factors and symptom recognition (Smyth et al., 2013; Woods, Moniz-Cook, & Iliffe, 2003). Community nurses encountered challenges when attempting to meet the needs of people with dementia in dementia management, social support, and pharmacological treatment (Eichler et al., 2016). Additionally, they demonstrated less positive attitudes and more negative emotions towards dementia (Lewis & Stenfert-Kroese, 2010), which could contribute to stress and turnover rate. Few dementia education programs were designed to suit the needs of community nurses in improving early detection and timely diagnosis of dementia in the community setting.

Most studies on continued education have focused on memory clinics, collaborative care, and case findings among primary care teams (Callahan et al., 2006; Kallumpuram et al., 2015; Lee et al., 2010). Although these studies showed positive results concerning community nurses' role in dementia care and subsequent management, they cannot be generalized to China and other LMICs because these countries lack government-funded services and the professional role of community nurses in dementia care is limited (Wang et al., 2014). Due to the lack of resources and time constraint, a cost-saving, sustainable, and acceptable program for community nurses is needed in these areas.

Web-based dementia training has become an important supplement for traditional education. It overcomes the lack of educators, time, and resource limitations of on-site education. It is reported that web-based dementia training programs for community nurses increased nurses' knowledge in communication (Chao et al., 2016), feeding skills (Batchelor-Murphy, Amella, Zapka, Mueller, & Beck, 2015), and attitude towards dementia and self-efficacy (Irvine, Beaty, Seeley, & Bourgeois, 2013) in HICs. However, dementia education programs require interactions with educators to develop community nurses' capability in dementia care, enable them to make changes in practice, and support them to overcome difficulties and barriers in their workplaces.

It is widely recognised that using smart phones can improve reach-out and interactions with population in need in health promotion and disease prevention (Dubey et al., 2014). However, using this platform to enhance interactions with health professionals in dementia care education is scarce. WeChat, a Chinese simple mobile applet, enables interactive learning between educators and learners. It is cost-saving, less resource-intensive, and more accessible than videos and online courses. It has no drawbacks among computer-required training, which is not available for staff lacking computer skills and undeveloped districts (Abdelaziz, Samer, Karam, & Abdelrahman, 2011). Moreover, the text, voice, video, and images generated on WeChat can be transmitted online without limitations of time and space. The WeChat-based approach has been widely used in clinical education (Zeng, Deng, Wang, & Liu, 2016), questionnaire collection (Sun, Zhu, Liang, Xu, & Lang, 2016), and health education and management (Kang et al., 2016). Although WeChat is designed in China, it shares similarities with many mobile applets used elsewhere. Evidence generated from the use of this kind of learning space for nurses in primary care will be valuable for the international community.

Although using WeChat to enhance the outcome of a dementia education program seems feasible, no study has tested its effectiveness in a dementia-specific education program with community nurses. Therefore, this study determined whether a dementia-specific education program that incorporated WeChat-based learning interactions could improve nurses' dementia knowledge, attitudes, and their intentions to make changes to achieve early detection and a timely diagnosis of dementia. It was hypothesized that nurses' dementia knowledge, attitudes, and intentions to make changes to achieve dementia early detection and a timely diagnosis would improve after dementia-specific training. Additionally, we expected that the program would be well-received by participants.

2. Methods

2.1. The dementia-specific training program for primary care settings

The authors of this study designed the training program for primary care nurses. We applied a 3-factor education intervention model (Green, Kreuter, & Deeds, 1980). This model emphasises (1) the use of lectures and written materials to disseminate learning content to learners; (2) developing resources to support learners to implement new skills and (3) reinforcing new skills by providing timely feedback and peer support (Aylward, Stolee, Keat, & Johncox, 2003). The learning content we developed was based on the analysis of learning needs of participants in the program through our discussions with nurse managers. We also analysed and incorporated the latest development of dementia care knowledge, skills, competencies and guidelines reported in the literature (Alzheimer's Disease International, 2011; Alzheimer's Disease International, 2012; Alzheimer's Disease International, 2014; Annear et al., 2015). The program included both lectures and WeChat learning interactions with educators and peers and self-directed learning activities. The lectures consisted of two modules focused on overall dementia understanding, early detection, and quality care strategies (Table 1). Module 1 included topics of 'understanding dementia', etiologies, types of dementia, clinical manifestation, risk factors and early detection. Module 2 included person-centred dementia care, cognitive function assessment and management of dementia associated symptoms and behaviours and support caregivers. The WeChat-based interaction provides an in-depth knowledge covered at lectures, which is enforced by online peer discussion.

2.2. The process to implement the program

The program team included a geriatric nursing specialist and three academic members who specialized in dementia care and psychology from a university in China. A 2-day workshop was conducted to review and discuss the preliminary topics based on the discussions with communities' nurse managers. The program included on-site lectures and WeChat online interactions. Nurse managers in CHSCs were invited to develop a timetable for the program to maximize nurses' attendance rates. Participants then received 2-weekly lectures provided by the three academic members in the team and each lecture lasted 3 h. The lecture slides were available online for participants to access when needed. The interactive learning via WeChat lasted 3 months and was facilitated by the geriatric nursing specialist. Learning activities via the Wechat included group discussions on how to apply knowledge to own practice, sharing experiences in dementia care and in overcoming barriers and difficulties in practice. The research team members provided ongoing support for the geriatric nursing specialist to deliver the WeChat-based learning activities though WeChat videos and telephone during the program implementation phase.

Table 1
Training modules.

Modules	Content
Module 1	Understanding dementia Etiologies Types of dementia Clinical manifestation Risk factors and early detection
Module 2	Person-centred dementia care Cognitive function assessment and management of dementia-associated symptoms and behaviours Support caregivers

2.3. Design and procedures

The study was a two-arm clustered randomized controlled trial in four community health services centers (CHSCs). Two were randomly designated as training facilities and another two were control facilities. This study was conducted from November 2015 to March 2016. The program's effects on the intervention group (IG) were examined in comparison to the control group (CG). Data were collected at baseline (T1), after 2 weeks (T2, immediately after lectures) and at a 3-month follow-up (T3, after the completion of the 3-month WeChat interaction).

2.4. Setting

The study was conducted in Chongqing, China, which has a population of 29.91 million. The proportion of those aged 60 years or older was 17.42% in 2010 (National Bureau of Statistics, 2012), which is the highest among China's 31 regions. The CHSC functions as primary institutions offering basic medical and public health services in China (Pan, Dib, Wang, & Zhang, 2006). Four CHSCs were randomly selected from a pool of 23 that provide services to 29 communities with a total population of 216,000.

2.5. Sample

Registered nurses (RNs) in primary care setting in Shapingba who had role and responsibilities in caring for older people were invited to participating in the program, and 115 RNs entered at baseline. The inclusion criteria were: (1) RNs with at least a 1-year working experience in primary care; (2) RNs with a role and responsibilities in caring for older people in primary care, (3) access to smartphone and Internet and (4) willingness to participate in the trial. The exclusion criteria were: (1) recruited in other dementia-related program; (2) nurses with less 1-year working experience; (3) RNs only working in child and maternal care areas in primary care. It is noted that RNs did not work in different sites, minimising the risk of cross-contamination between control and training sites. Verbal, informed consent was obtained from participants and the data was treated confidentially and anonymously. The ethical committee of Third Military Medical University granted ethical approval and this study was conducted in accordance with the Declaration of Helsinki.

A power calculation was performed to determine sample size. To detect a high effect size ($f = 0.55$) at a significant level of 5% and 80% power using an analysis of variance (ANOVA), at least 30 participants were required in each group (Lenth, 2006–2009). Therefore, the target recruitment was 36 nurses in each group considering a 20% attrition rate.

2.6. Conditions

2.6.1. IG

Participants in the IG received dementia-specific training. Nurses initially received 2-weekly lectures lasting three-hours each with a face-to-face discussion, which was delivered by a geriatric nurse specialist with training slides and handouts given to participants for reference. Thereafter, they were enrolled a WeChat group online interaction, which continued for three months and comprised videos, case studies, readings, and so on. Nurses discussed the materials and the geriatric specialist facilitated the discussions.

2.6.2. CG

Nurses in the CG received training for the care of older people with disability. They received 2-weekly lectures lasting three-hours each with a face-to-face discussion. A geriatric nurse specialist delivered the lectures and handouts and slides were given to participants. The CG also received the dementia-specific training following the completion of research participation.

2.7. Measures

Two validated and one self-developed questionnaires were used to assess nurses' knowledge and attitudes towards dementia, and their training acceptance. The intentions to make changes to achieve early detection and a timely diagnosis of dementia were measured by one self-developed, close-ended question.

2.7.1. Chinese Alzheimer's Disease Knowledge Scale (CADKS)

The CADKS was utilized to assess dementia knowledge. It consists of 30 true/false items; total scores range from 0 to 30. The items include 7 content domains: risk factors, symptoms, disease course, assessment and diagnosis, treatment and management, life impact, and caregiving. The scale was originally designed in 2009 and had an overall internal consistency of Cronbach's $\alpha = 0.71$ (Carpenter, Balsis, Otilingam, Hanson, & Gatz, 2009). The CADKS showed good content validity (0.91) and internal consistency in this study (Cronbach's $\alpha = 0.72$) (Wang, Xiao, & He, 2015).

2.7.2. Chinese Dementia Attitudes Scale (CDAS)

The CDAS is a 7-point Likert scale comprising 20 items that was developed for students and care workers. The total score ranges from 20 to 140. It contains two subscales: dementia knowledge and social comfort. Higher scores suggest more positive attitudes towards people with Alzheimer's disease and related dementias (ADRD). The original version was developed to reflect the affection, behavior, and cognitive reactions towards people with ADRD. It showed good reliability (Cronbach's $\alpha = 0.84$) (O'Connor & McFadden, 2010). The CDAS was translated by the research team from the original version of the DAS. The CDAS had a high internal consistency (Cronbach's $\alpha = 0.79$) in this study.

2.7.3. Intentions to make changes to achieve early detection and a timely diagnosis of dementia

Nurses' intentions to make changes to achieve early detection and a timely diagnosis of dementia were measured by one self-developed, closed-ended question, 'How do you cope with a client with subjective memory complaints'? The choices consisted of 'advice for cognitive screening', 'perform cognitive screening for clients', 'suggest to see specialists', 'comfort clients', 'ignore it', and 'others'. The responses were collected at T1 and T3. The preferred answers were 'advice for cognitive screening', 'perform cognitive screening for clients' and 'suggest to see specialists'.

2.7.4. Program acceptance

Program acceptance was assessed using a self-developed, training satisfaction questionnaire and written feedback. The 10-item satisfaction questionnaire was developed from an Internet training program literature and was measured using a 5-point Likert scale (Irvine et al., 2013). Scores ranged from 10 to 50; higher scores indicated higher satisfaction. One open-ended question was added asking nurses to write comments on the program's positive aspects and suggestions for improvement. Participants completed the questionnaire and written feedback at T2 and T3.

2.8. Statistical analysis

Data was analysed using SPSS 19.0 (SPSS, Chicago, IL, USA). Analysis was conducted based on intention-to-treat (ITT) based on patients' assignment regardless of the change of site after the assignment. (Sundram, Dahlui, & Chinna, 2014). The Kolmogorov-Smirnov test was used to determine the normality distribution. Normally distributed descriptive data were reported with mean scores and standard deviations, and the skewed distributed as median with 25th (Q1) and 75th percentiles (Q3). Demographic characteristics were compared using a *t*-test for normally distributed continuous variables, the Mann-Whitney *U* test for continuous variables with skewed distributions, and the χ^2 test for categorical variables. A non-parametric test was used if the normality was violated. The *p*-value was set at 0.05 for statistical significance.

For the CADKS and CDAS data, a two-sample *t*-test was used to compare the baseline data of the IG and CG, and a repeated measures ANOVA was used to assess differences between the groups' changes over time. If the ANOVA showed a significant group × time interaction, a two-sample *t*-test was used to compare the data of the two groups at T2 and T3 with mean difference (MD) and a 95% confidence interval (CI) provided.

3. Results

3.1. Participants' characteristics

One-hundred fifteen participants were enrolled, and 101 (valid response rate = 87.8%) completed all three questionnaire assessments. Ten nurses discontinued at T2 due to a lack of time to attend the second lecture, and four nurses resigned at T3. Nurses who did not continue did not differ significantly from those who continued in baseline characteristics (Fig. 1). Demographic details for both groups and comparisons between groups are summarized in Table 2. No significant differences were found between groups in baseline characteristics.

3.2. Knowledge

As shown in Table 3, an ANOVA was chosen to assess difference between groups in changes over time for total CADKS and its 7 subscales. For CADKS, a significant group × time interaction ($F = 31.35$, $p < 0.001$) was found; *t*-tests indicated that compared to the CG, the IG showed higher scores at T2 and T3.

Regarding the subscales, neither a significant group × time interaction ($F = 0.75$, $p = 0.465$) nor significant main effects of group ($F = 3.96$, $p = 0.065$) were found in 'life impact', indicating that there was no significant difference in change between groups. For 'care giving' and 'assessment and diagnosis', no significant group × time interactions ($F = 0.87$, $p = 0.421$; $F = 0.41$, $p = 0.657$, respectively) were found; however, significant main effects of groups were observed ($F = 14.26$, $p = 0.005$; $F = 14.20$, $p < 0.001$, respectively), indicating that there were significant differences in changes for these two factors. For 'risk factors', 'symptoms', 'disease course', and 'treatment and management', significant group × time interactions ($F = 16.63$, $p < 0.001$; $F = 10.52$, $p < 0.001$; $F = 14.40$, $p < 0.001$; $F = 7.99$, $p < 0.001$, respectively) were detected. Therefore, *t*-tests were performed to assess the follow-up differences between groups, and the results showed significant differences between groups at T2 and T3.

3.3. Attitudes

Table 4 shows the changes of attitudes scores between groups over time. An ANOVA revealed significant group × time interactions for total CDAS score ($F = 20.57$, $p < 0.001$), 'social comfort' ($F = 16.70$, $p < 0.001$) and 'dementia knowledge' ($F = 5.85$, $p = 0.004$). *t*-tests were performed to assess follow-up differences between groups, and the result showed significant differences between groups at T2 and T3.

3.4. Intentions to make changes to achieve early detection and a timely diagnosis of dementia

Most participants (86.1%) encountered clients' subjective memory complaints. Table 5 presents the two groups' recommendations for clients with subjective memory complaints. At T1, no significances were found among responses, whereas the CG endorsed a greater proportion of 'suggest to see specialists' than the IG did, albeit not significant ($p = 0.200$). Following the training, more participants in the IG 'gave advice of cognitive screening' ($p = 0.025$) and 'perform cognitive screening for clients' ($p = 0.015$) than participants in the CG did. However, 'suggest to see specialists' did not show difference between the two groups ($p = 0.200$).

3.5. Program acceptance

Table 6 displays nurses' program satisfactory scores. Each item was rated > 4 score at T2, indicating a high acceptance level. The satisfaction of each item was maintained at T3, with item two ('learned a lot'), three ('risk factors and preventive strategies'), and nine ('more efficient patient care') revealing significant decreases ($p = 0.007$, $p = 0.041$, and $p = 0.048$, respectively). Global satisfaction decreased significantly from T2 to T3 ($p = 0.035$).

The written feedback provided by 16 (valid response rate = 32.7%) participants at T2 identified several positive aspects: the content was

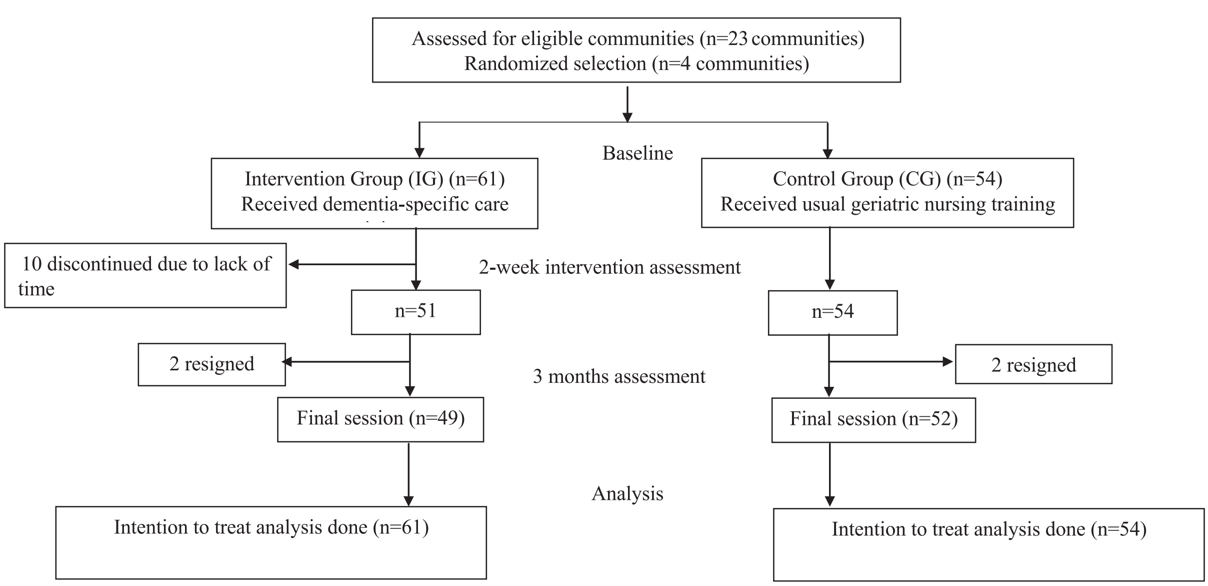

Fig. 1. Participant flow chart.

Table 2
Descriptive statistics of study participants.

Variable	Total (n = 115)	IG (n = 61)	CG (n = 54)	P-value
Gender				
Female, n (%)	108 (93.9)	57 (93.4)	51 (94.4)	0.823[b]
Age, Mean (SD)	30.75 (9.17)	28.8 (8.97)	31.9 (8.69)	0.068[a]
Education level, n (%)				0.352[b]
Trade school	26 (2.6)	11 (18.0)	15 (27.8)	
Some college	64 (55.7)	34 (55.7)	30 (55.6)	
College or professional graduate	25 (21.7)	16 (26.2)	9 (16.37)	
Overall clinical experience, Median (Q1; Q3)	7.00 (3.00;16.00)	5.00 (3.00;12.50)	7.00 (3.75;11.00)	0.059[c]
Community work experience, Median (Q1; Q3)	4.00 (2.00;8.00)	3.00 (2.00;6.00)	6.00 (2.00;9.25)	0.087[c]
Employment status, n (%)				0.346[b]
Part-time	81 (70.4)	45 (73.8)	36 (66.7)	
Full-time	34 (29.6)	16 (26.2)	18 (33.3)	
Prior dementia contact (yes), n (%)	75 (65.2)	39 (63.9)	36 (66.7)	0.759[b]
Prior dementia care (yes), n (%)	28 (24.3)	18 (29.5)	10 (18.5)	0.171[b]
Clients' subjective memory complaints (yes), n (%)	99 (86.1)	54 (88.5)	45 (83.3)	0.422[b]
Prior dementia training (yes), n (%)	43 (37.4)	24 (39.3)	19 (35.2)	0.645[b]
Willingness of dementia learning (yes), n (%)	109 (94.8)	59 (96.97)	50 (92.6)	0.320[b]
WeChat online learning (yes), n (%)	42 (36.5)	19 (31.1)	23 (42.6)	0.203[b]

Note: IG = Intervention Group; CG = Control Group.
[a] *t*-test.
[b] χ^2 test.
[c] Mann-Whitney *U* test.

Table 3
Results of ANOVA analyses for CADKS: differences between groups in change scores from baseline to 2-week and 3-month follow-up.

Items	Mean (SD)		t^a (P)	F^b_g (P)	F^c_t (P)	$F^d_{t \times g}$ (P)	t^e (P)	MD (95%CI)
	IG (n = 61)	CG (n = 54)						
Total CADKS				41.62 (< 0.001)	63.08 (< 0.001)	31.35 (< 0.001)		
T1	19.43 (2.48)	19.04 (2.66)	0.81 (0.42)					
T2	24.18 (3.51)	20.02 (2.53)					7.21 (< 0.001)	4.16 (3.02–5.31)
T3	23.61 (3.37)	19.54 (2.67)					7.11 (< 0.001)	4.07 (2.94–5.20)
Risk factors				31.41 (< 0.001)	43.27 (< 0.001)	16.63 (< 0.001)		
T1	3.59 (1.16)	3.44 (1.13)	0.68 (0.49)					
T2	5.11 (1.15)	3.81 (0.89)					6.69 (< 0.001)	1.30 (0.92–1.68)
T3	5.00 (1.11)	3.76 (1.13)					5.93 (< 0.001)	1.24 (0.83–1.66)
Symptoms				6.70 (0.011)	14.15 (< 0.001)	10.52 (< 0.001)		
T1	2.03 (0.98)	2.19 (0.99)	0.83 (0.41)					
T2	3.00 (1.09)	2.26 (0.89)					3.94 (< 0.001)	0.74 (0.37–1.11)
T3	2.67 (0.91)	2.22 (0.72)					2.92 (0.004)	0.45 (0.15–0.76)
Course of the disease				11.13 (0.001)	14.99 (< 0.001)	14.40 (< 0.001)		
T1	2.66 (0.94)	2.72 (0.92)	0.38 (0.70)					
T2	3.38 (0.88)	2.87 (0.70)					3.39 (0.001)	0.51 (0.21–0.80)
T3	3.46 (0.85)	2.63 (0.81)					5.35 (< 0.001)	0.83 (0.52–1.14)
Assessment and diagnosis				14.20 (< 0.001)	0.69 (0.499)	0.41 (0.657)		
T1	3.07 (0.77)	2.80 (0.92)	1.71 (0.09)					
T2	3.21 (0.73)	2.78 (0.72)						
T3	3.08 (0.74)	2.69 (0.84)						
Treatment and management				5.94 (0.016)	7.15 (0.001)	7.99 (< 0.001)		
T1	2.95 (0.78)	3.09 (0.68)	1.03 (0.31)					
T2	3.57 (0.62)	3.11 (0.92)					3.19 (0.002)	0.46 (0.18–0.75)
T3	3.41 (0.67)	2.98 (0.86)					3.01 (0.003)	0.43 (0.15–0.71)
Life impact				3.96 (0.065)	0.02 (0.97)	0.75 (0.465)		
T1	2.30 (0.64)	2.20 (0.60)	0.79 (0.43)					
T2	2.34 (0.54)	2.15 (0.63)						
T3	2.36 (0.58)	2.11 (0.74)						
Care giving				14.26 (0.005)	22.46 (< 0.001)	0.87 (0.421)		
T1	2.84 (0.97)	2.59 (0.94)	1.36 (0.18)					
T2	3.54 (1.06)	3.04 (1.06)						
T3	3.62 (0.98)	3.15 (1.04)						

Note: IG = Intervention Group; CG = Control Group; T1 = Baseline; T2 = 2-week post-baseline; T3 = 3-month follow-up.
[a] Two sample *t*-test was used to compare the baseline data of two groups.
[b] Repeated measures analysis of variance (ANOVA), F_g means main effect of the group.
[c] Repeated measures analysis of variance (ANOVA), F_t means main effect of time.
[d] Repeated measures analysis of variance (ANOVA), $F_{t \times g}$ means group × time interaction.
[e] If the result of ANOVA showed significant group × time interaction, two sample *t*-test was used to compare the data of two groups at T2 and T3.

Table 4
Results of ANOVA analyses for CDAS: differences between groups in change scores from baseline to 2-week and 3-month follow-up.

Items	Mean (SD)		t^a (P)	F^b_g (P)	F^c_t (P)	$F^d_{t \times g}$ (P)	t^e (P)	MD (95%CI)
	IG (n = 61)	CG (n = 54)						
Total CDAS				29.45 (< 0.001)	22.28 (< 0.001)	20.57 (< 0.001)		
T1	89.08 (13.70)	87.35 (11.03)	0.74 (0.46)					
T2	103.34 (14.65)	88.43 (12.03)					5.92 (< 0.001)	14.92 (9.93–19.91)
T3	103.77 (16.53)	86.93 (12.07)					6.17 (< 0.001)	16.85 (11.44–22.25)
Social comfort				22.19 (< 0.001)	16.78 (< 0.001)	16.70 (< 0.001)		
T1	37.59 (8.63)	37.24 (7.16)	0.23 (0.81)					
T2	44.57 (9.50)	37.11 (8.11)					4.50 (< 0.001)	7.46 (4.18–10.75)
T3	47.48 (10.60)	37.30 (7.40)					5.90 (< 0.001)	10.18 (6.76–13.60)
Dementia knowledge				15.38 (< 0.001)	9.64 (< 0.001)	5.85 (0.004)		
T1	52.49 (11.24)	50.11 (9.80)	0.70 (0.49)					
T2	58.77 (8.06)	51.31 (7.55)					5.10 (< 0.001)	4.56 (6.93–10.36)
T3	56.30 (9.36)	49.63 (8.78)					3.92 (< 0.001)	6.67 (3.30–10.03)

Note: IG = Intervention Group; CG = Control Group; T1 = Baseline; T2 = 2-week post-baseline; T3 = 3-month follow-up.
[a] Two sample *t*-test was used to compare the baseline data of two groups.
[b] Repeated measures analysis of variance (ANOVA), F_g means main effect of the group.
[c] Repeated measures analysis of variance (ANOVA), F_t means main effect of time.
[d] Repeated measures analysis of variance (ANOVA), $F_{t \times g}$ means group × time interaction.
[e] If the result of ANOVA showed significant group × time interaction, two sample *t*-test was used to compare the data of two groups at T2 and T3.

Table 5
Intentions to make changes to achieve early detection and a timely diagnosis of dementia.

Items	IG (n = 61)		CG (n = 54)		P-value [T1]	P-value [T3]
	T1 (n, %)	T3 (n, %)	T1 (n, %)	T3 (n, %)		
1. Advice for cognitive screening	18, 29.5	43, 70.5	13, 22.2	27, 50.0	0.512	0.025
2. Perform cognitive screening for clients	10, 16.4	28, 45.9	5, 9.3	13, 24.1	0.257	0.015
3.Suggest to see specialists	18, 29.5	30, 49.2	24, 44.4	33, 61.1	0.097	0.200
4.Comfort clients or ignore the topic	15, 24.6	3, 4.9	13, 24.1	3, 5.6	0.949	0.878

Note: IG = Intervention Group; CG = Control Group; P-value [T1] = IG/CG; P-value [T3] = IG/CG.T1 = Baseline; T3 = 3-month follow-up.

informative, easy to follow, and useful for nurses' personal abilities and health education in the future; nurses learned many dementia care and communication strategies; and the training was a refresher of knowledge and good practice. Ten (20.4%) nurses stated needing more clinical case studies, up-to-date research on Alzheimer's disease, and more dementia video examples and WeChat interactions, which were concise and easy to comprehend. Nine nurses suggested using similar programs in the future.

At T3, 13 (26.5%) participants reported effective communication with dementia patients and caregivers, endorsed more daily care strategies, and more cognitive screening for clients using the 'clock drawing test' or the Mini Mental State Examination (MMSE). Eight nurses reported increased understanding of BPSD and suggested the use of more dementia case studies, detailed dementia daily care, and WeChat interactions.

4. Discussion

The program increased nurses' overall dementia knowledge immediately after lectures, and the improvement was maintained at the 3-month follow-up. This means the WeChat-based short-term training program could enhance nurses' knowledge. The result echoed the results of an on-site dementia training program that improved dementia knowledge in hospitals (Elvish et al., 2014), nursing homes (Irvine et al., 2013), and an informative website program for lay individuals (Hughes, Lowe, Shine, Carpenter, & Balsis, 2015). In addition, the WeChat-based self-directed training format echoed the result of a self-directed, learning package investigation that enhanced rural aged palliative care workers' knowledge (Pitman, 2013). Although the baseline CADKS was lower than it was in other countries (Smyth et al., 2013), nurses' knowledge increased through this brief and effective training, which means the program met nurses' needs.

Regarding reinforcing factors, the continued WeChat-based sharing of dementia information and discussion with peers may contribute to the post-test improvement and maintenance. Per Adult Learning Theory, personal desires of learning enable individuals to understand or

Table 6
Scores of training satisfaction (n = 52).

Items	T2 Median (Q1;Q3)	T3 Median (Q1;Q3)	P-value
1. I can apply the training to my daily work.	4 (4;5)	4 (4;5)	0.291
2. I learned a lot from the training.	5 (4;5)	4 (4;5)	0.007
3. Dementia risk factors and preventive strategies are very useful to me.	5 (4;5)	4 (4;5)	0.041
4. Dementia screening strategies and early recognition knowledge are very useful to me.	5 (4;5)	4 (4;5)	0.088
5. The understanding and management of BPSD of the dementia people are very useful to me.	5 (4;5)	4 (4;5)	0.180
6. The communication strategies and daily care of dementia people are very useful to me.	5 (4;5)	4 (4;5)	0.695
7. After the face-to-face lectures, the knowledge from the online learning materials are very useful to me.	5 (4;5)	4 (4;5)	0.273
8. Compared to similar dementia training, I learned more using an online WeChat interaction.	4 (4;5)	4 (4;5)	0.747
9. This training made my work more efficient, I will be able to provide better care for the patients.	5 (4;5)	4 (4;5)	0.048
10. This training made my work more efficient, I feel more satisfied with my job.	4 (4;5)	4 (4;5)	0.217
Total score	46 (40;50)	43 (40;48)	0.035

Note: T2 = 2-week post-baseline; T3 = 3-month follow-up.

perform a job more effectively, thus inspiring a higher motivation to learn (Knowles, 1996). Considering the subscales, the IG presented a significant improvement of 'risk factors', which is rarely reported and considered crucial for dementia risk reduction in primary care teamwork. Additionally, mastery of dementia risk factors is not only essential for dementia early detection, referral behaviours in practice, and quality dementia care in public health services, but is also helpful for general population health and caregivers.

Results showed that there was no significant change over time in 'life impact' between groups. One possible reason is that a ceiling effect may have left little scope for change as the scores of the three items were relatively good at baseline. This result implies that most nurses may have good knowledge of dementia patients' daily life because 65.2% of participants had prior experience working with patients with dementia. In addition, no significant interaction effect was found in 'care giving' or 'assessment and diagnosis'. There are possible reasons for this. First, most nurses lacked dementia care experience, as only 24.3% of the participants had previous dementia care experience. Second, nurses implemented minimal dementia assessment and diagnosis in clinical practice as they had a restricted professional role in clinical practice and dementia was not included in the government pharmaceutical benefit scheme (Ministry of Human Resources and Social Security, 2009). Nonetheless, the significant differences between groups of these two aspects indicated that the program was still effective in enhancing nurses' dementia knowledge.

The findings showed that nurses' attitudes concerning dementia improved over time, reflecting a positive inclination of dementia quality care. This finding is similar to a previous study, which reported a positive effect of DAS on nursing students' attitudes (George, Stuckey, & Whitehead, 2013); however, the baseline level in the present study was lower, which may be caused by an unequal allocation of healthcare resources and inaccessibility of learning programs for Chinese nurses (Xiao, 2010). However, the results contrast with those of Chao et al. (2016), who reported no communication attitudes changes of nurses towards people with dementia, which may be due to the participants' characteristic and the intervention content. Nurses' improved dementia attitudes is considered important in reducing dementia stigma and enabling people with dementia to acknowledge their symptoms, their need for medical treatment, and help them achieve a high quality of life (Alzheimer's Disease International, 2012).

Notably, the CDAS' subscales 'dementia knowledge' and 'social comfort' mainly test attitude and indirectly measures nurses' knowledge-related attitude towards dementia; therefore, the low baseline attitude level may have been partially caused by the lack of dementia knowledge, which emphasized the need for a dementia training program to eliminate the knowledge-induced stigma. It has been reported that attitudes are central to ones' being and are highly resistant to change (Petty, Tormala, Brinol, & Jarvis, 2006). However, our study showed a positive effect of attitudes by a convenient and efficient approach, WeChat, which provides users a boundary-free platform for access to information access based on personal interests, needs, and pace. Additionally, this platform promotes online scholarly discussions and continued exchange of knowledge and experience. This is consistent with other web-based or smartphone software programs such as Dementia Pathway Project (Ollerenshaw, 2015), iVitality (Van Osch et al., 2015) and eButton (Sun et al., 2014) reported from other countries to assist primary health practitioners to improve dementia diagnosis, referral and management, dementia home-based health monitoring and personal assistance. Despite the differences of healthcare system among countries in the world, WeChat applet along with other smartphone applets provides a new platform to improve reach-out in dementia education for health professionals in primary care, especially those working in rural and remote areas (Hilty et al., 2013).

Another priority for the current investigation was to explore the influence of training on nurses' intentions to make changes to achieve early detection and a timely diagnosis of dementia. In China, < 10% of people with dementia were correctly diagnosed (Chen et al., 2013), and the diagnosis was delayed for approximately 2 years even in well-established memory clinics (Zhao et al., 2016). The nature of dementia early detection is influenced by nurses' knowledge of dementia and the understanding of their professional role (Trickey et al., 2000). In the current study, the IG demonstrated significantly improved intention to change practice to achieve early detection and a timely diagnosis compared with the CG. Nurses reported the use of 'clock drawing test' and MMSE in cognitive assessment for clients. The post-test improvement may be due to the limited baseline response and their enthusiasm of continued education; moreover, the training content catered to their needs. In addition, the on-going WeChat interaction facilitated by specialists can further strengthen their practice consistently. Nurse-led case finding for suspected dementia in high-risk groups is considered important in maximizing decision autonomy and enabling earlier access to information, support, and medical treatment (Kallumpuram et al., 2015). Moreover, well-trained nurses could also raise public awareness of dementia through health education, which could lead to an increase of people with memory problems being referred to memory clinics. Although no significant difference was found in 'suggest to see specialists' in the two groups at the 3-month follow-up, the overall results suggest that a brief, intensive program combined on-site and utilizing a WeChat format was feasible and well-received by nurses. This will increase intentions to make changes to achieve early detection and a timely diagnosis of dementia and the use of formal validated cognitive tests.

As expected, a high level of satisfaction was identified in this study. This was consistent with previous research that revealed that students preferred 'blended learning', which integrates both strengths of e-learning and lecture (Abdelaziz et al., 2011). However, our study showed a significant decrease of satisfaction from T2 to T3 due to the second ('learned a lot'), third ('risk factors and preventive strategies') and ninth ('more efficient patient care') items.

There are several possible explanations for this results. First, some nurses' learning needs were not fully met as they required more case studies and dementia care strategies (Page & Hope, 2013). Second, nurses' limited professional role, busy and stressful clinical practice, and low salaries worsened their enthusiasm in dementia care (Wei, Zakus, Liang, & Sun, 2005). In this study, four nurses resigned because of low salary, stressful work, and occupational burnout. Moreover, they preferred to work in comprehensive hospitals rather than in primary care settings. Third, the transfer of knowledge to practice is difficult to maintain new practices in the complex organizational cultures of hospitals (Waugh, Marland, Henderson, Robertson, & Wilson, 2011). Despite the significant decreased satisfactory scores, every item still fell within the 'agree' category, indicating the positive training effect was sustained at the 3-month follow-up. In addition, participation was stable and attrition was low, with 101 (87.8%) of an initial 115 completing the follow-up. The high adherence level supported the feasibility of this training format and intervention strategies in future studies. The findings suggest that more financial support to dementia care and management, more dementia-related training targeted primary care staff, and more professional autonomy of this group are needed.

Overall, nurses reported positive feedback regarding the program. At T2, nurses indicated that they learned practical strategies relevant to clinical practice. Suggestions for improvements highlighted that they required more case studies, video examples, online interaction, and Alzheimer's disease frontier research materials. At T3, responses revealed that the strategies were both helpful and could be used in practice. Moreover, they suggested that more detailed dementia care skills and hands-on guidance are necessary, which explained their decreased satisfaction scores and indicated why trainings with different depth have an impact on satisfaction (Surr, Smith, Crossland, & Robins, 2016). Based on nurses' suggestions, further studies could expand the current focus to include dementia care strategies more explicitly, and provide peer-support activities in addition to the WeChat online interaction.

This study had several limitations. First, the broader applicability of study findings may be affected as the program was conducted in a limited region in China (metropolitan Chongqing). The outcomes could have been affected if we had enrolled more participants from different districts with various background characteristics. Second, this program has limits regarding real life practice change through a self-report questionnaire. Future reports could include measures of staff interaction with dementia patients to assess the application of their knowledge gains. Third, it is difficult determining which part of intervention contributed to the effectiveness due to the use of combination of face-to-face and WeChat educational intervention. However, this study has established preliminary effectiveness of a brief, less resource-intensive intervention through a WeChat format, which is important for LMICs in reducing environmental barriers such as limited time, staff scheduling, and cost. Our next steps are to disseminate the dementia education program in different regions using WeChat and to evaluate its impact on people with dementia.

5. Conclusion

This study suggests that a WeChat-based dementia-specific training is effective in improving nurses' knowledge and attitudes towards dementia as well as their intentions to make changes to achieve dementia early detection and a timely diagnosis in primary care settings. Moreover, the program was feasible and well-accepted by nurses. More research is clearly needed, however, addressing how best to implement this type of adoption across settings.

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Conflicts of interest

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